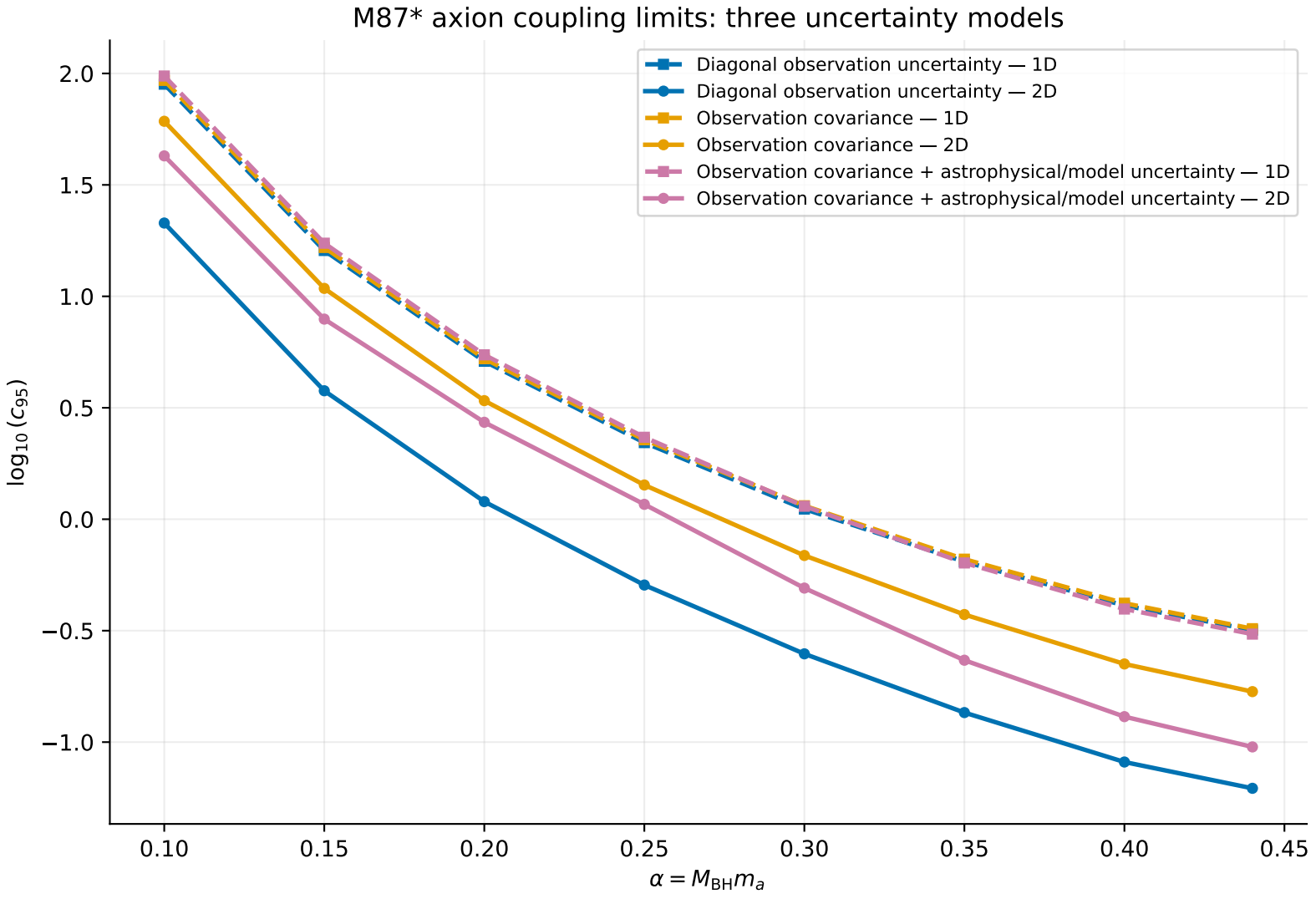
M87* axion coupling limits: three uncertainty models
Diagonal observation uncertainty — 1D
Diagonal observation uncertainty — 2D
Observation covariance — 1D
Observation covariance — 2D
Observation covariance + astrophysical/model uncertainty — 1D
Observation covariance + astrophysical/model uncertainty — 2D
$\log_{10}(c_{95})$
$\alpha = M_{\mathrm{BH}} m_a$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45